

ECE 410. Codefest 7 CMAN

1. a) $\text{Compute}_{\text{mvm}} = N^2$
 $= 512^2 \text{ FLOPs}$

b) $\text{Memory}_{\text{mvm}} = N^2(4)$
 $= 512^2(4) \text{ bytes}$

c) $\text{Compute}_{\text{sparse}} = (5 \cdot N) \cdot N^2$
 $= (5 \cdot 512) \cdot 512^2$

d) $\text{Memory}_{\text{sparse}} = 5(N(NAN)) \cdot (1(4)) \cdot 4$

2. $\text{speedup} = \frac{N^2}{(5N) \cdot N^2}$

$2 = \frac{N^2}{5N^2}$

$5 = \frac{\sqrt{2}}{2} \approx 0.707$

3. $512^2(4) = 4((5 \cdot 512(512)) \cdot 2^2) + (512^2) \cdot 4$

$512^2 = 2(512(512)) \cdot 2 + 1$

$512 = 2(512) \cdot 2 + 1$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

$512 = 5$

4. $\text{mem} = 0.9(512(512))(4)$
 $= 945562 \text{ bytes} < 320 \text{ GB} \checkmark$

$\text{speedup} = \frac{512^2}{(0.9512)^2}$
 $= 1.23457$